# x.keymap

7 layers · 34 keys · 3 combos

every arrow starts on the key that triggers it and ends on the layer name

**how to read**

- pink key = switches to another layer
- red key = held down while on this layer
- solid arrow = hold
- dashed arrow = toggle / go to / one-shot
- combo = press the red keys together
- arrow colour = the layer it leads to

## 0 default_layer — base layer

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Q | W | E | R | T | Y | U | I | O | P |
| A | S | D | F | G | H | J | K | L | ; |
| Z | X | C | V | B | N | M | , | . | / |
| | | | extend-L1 | Space | Shift | numbers-R2 | | | |

## 5 mirrorNumpad-R1+R2

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| + | 7 | 8 | 9 | ÷ | Tab | 🔊+ | ☀+ | ⏭ | Esc |
| Enter | 4 | 5 | 6 | × | Space | 🔊− | ☀− | ⏮ | mouse-L2+R1 toggle |
| . | 1 | 2 | 3 | − | RESET | 🔇 | | ⏯ | mirror Numpad-R1+R2 toggle |
| | | | 0 | Bksp | | | | | |

## 1 extend-L1

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Esc | ` | ~ | Sticky Shift | Tab | Del | - | GUI+0 | = | Bksp |
| Sticky Ctrl | Sticky Alt | Sticky Win | Space | Caps | ← | ↓ | ↑ | → | Enter |
| GUI+Z | GUI+X | GUI+C | GUI+V | GUI+L | 1 | 2 | 3 | 4 | 5 |
| | | | numpad-L1+L2 toggle | | | function-L1+R2 | | | |

## 4 numpad-L1+L2

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Esc | ⏭ | ☀+ | 🔊+ | Tab | ÷ | 7 | 8 | 9 | + |
| mouse-L2+R1 toggle | ⏮ | ☀− | 🔊− | Space | × | 4 | 5 | 6 | Enter |
| numpad-L1+L2 toggle | ⏯ | | 🔇 | RESET | − | 1 | 2 | 3 | . |
| | | | | | Bksp | 0 | | | |

## 3 function-L1+R2

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| F1 | F2 | F3 | F4 | F5 | F6 | F7 | F8 | F9 | F10 |
| F16 | F17 | F18 | F19 | F20 | F11 | F12 | F13 | F14 | F15 |
| F21 | F22 | F23 | F24 | | BT 0 | BT 1 | BT 2 | | |

## 6 mouse-L2+R1

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Scroll ← | Scroll ↓ | Move ↑ | Scroll ↑ | Scroll → | Scroll ← | Scroll ↓ | Move ↑ | Scroll ↑ | Scroll → |
| Esc | Move ← | Move ↓ | Move → | Shift | Shift | Move ← | Move ↓ | Move → | Esc |
| mouse-L2+R1 toggle | Alt | Btn 5 | Btn 4 | 🖱 | 🖱 | Btn 4 | Btn 5 | Alt | mouse-L2+R1 toggle |
| | | | 🖱 | 🖱 | 🖱 | 🖱 | | | |

## 2 numbers-R2

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 |
| ' | { | [ | ( | \ | - | = | ↑ | Esc | Enter |
| " | } | ] | ) | \| | Del | Bksp | ↓ | ← | → |
| | | | function-L1+R2 | Shift | mouse-L2+R1 toggle | | | | |

combo hold · hold · combo toggle · toggle · hold · toggle · toggle · hold

## combos — press the red keys together

- mirrorNumpad-R1+R2 (hold) — mirrorNumpad-layer
- numpad-L1+L2 (hold) — numpad-layer
- mouse-L2+R1 (toggle) — mouse-layer

# x.keymap

**keymap source, comments removed (bindings re-aligned as keyboard rows)**

```
#define ZMK_POINTING_DEFAULT_MOVE_VAL 1000
#define ZMK_POINTING_DEFAULT_SCRL_VAL 20

#include <behaviors.dtsi>
#include <dt-bindings/zmk/bt.h>
#include <dt-bindings/zmk/keys.h>
#include <dt-bindings/zmk/pointing.h>

&sk { ignore-modifiers; };

&mmv {
    time-to-max-speed-ms = <400>;
    acceleration-exponent = <1>;
    delay-ms = <0>;
};

&msc {
    time-to-max-speed-ms = <300>;
    acceleration-exponent = <0>;
    delay-ms = <0>;
};

/ {
    macros {
        unstick: unstick {
        label = "ZM_unstick";
        compatible = "zmk,behavior-macro";
        #binding-cells = <0>;
        bindings = <&kp LSHIFT &kp RSHIFT &kp LCTRL &kp RCTRL &kp LALT &kp RALT &kp LGUI &kp RGUI>;
    };
};

combos {
    compatible = "zmk,combos";

    mirrorNumpad-layer {
        timeout-ms = <250>;
        key-positions = <32 33>;
        bindings = <&mo 5>;
    };

    numpad-layer {
        timeout-ms = <250>;
        key-positions = <30 31>;
        bindings = <&mo 4>;
    };

    mouse-layer {
        bindings = <&tog 6>;
        key-positions = <31 32>;
        timeout-ms = <51>;
    };
};

keymap {
    compatible = "zmk,keymap";

    default_layer {
    bindings = <
    &kp Q   &kp W   &kp E   &kp R   &kp T            &kp Y           &kp U   &kp I       &kp O     &kp P
    &kp A   &kp S   &kp D   &kp F   &kp G            &kp H           &kp J   &kp K       &kp L     &kp SEMICOLON
    &kp Z   &kp X   &kp C   &kp V   &kp B            &kp N           &kp M   &kp COMMA   &kp DOT   &kp SLASH
                            &mo 1   &kp SPACE        &kp LEFT_SHIFT  &mo 2

    >;
};

extend-L1 {
    bindings = <
    &kp ESC            &kp GRAVE      &kp TILDE      &sk LSHIFT   &kp TAB              &kp DELETE   &kp MINUS   &kp LG(N0)   &kp EQUAL   &kp BACKSPACE
    &sk LEFT_CONTROL   &sk LEFT_ALT   &sk LEFT_WIN   &kp SPACE    &kp CAPSLOCK         &kp LEFT     &kp DOWN    &kp UP       &kp RIGHT   &kp ENTER
    &kp LG(Z)          &kp LG(X)      &kp LG(C)      &kp LG(V)    &kp LG(L)            &kp N1       &kp N2      &kp N3       &kp N4      &kp N5
                                                     &trans       &tog 4               &none        &mo 3

    >;
};

numbers-R2 {
    bindings = <
    &kp N1              &kp N2            &kp N3              &kp N4                  &kp N5                &kp N6       &kp N7          &kp N8     &kp N9     &kp N0
    &kp SQT             &kp LEFT_BRACE    &kp LEFT_BRACKET    &kp LEFT_PARENTHESIS    &kp BACKSLASH         &kp MINUS    &kp EQUAL       &kp UP     &kp ESC    &kp ENTER
    &kp DOUBLE_QUOTES   &kp RIGHT_BRACE   &kp RIGHT_BRACKET   &kp RIGHT_PARENTHESIS   &kp PIPE              &kp DELETE   &kp BACKSPACE   &kp DOWN   &kp LEFT   &kp RIGHT
                                          &mo 3                &kp LEFT_SHIFT         &tog 6       &trans

    >;
};

function-L1+R2 {
    bindings = <
    &kp F1    &kp F2    &kp F3    &kp F4    &kp F5          &kp F6         &kp F7         &kp F8         &kp F9    &kp F10
    &kp F16   &kp F17   &kp F18   &kp F19   &kp F20         &kp F11        &kp F12        &kp F13        &kp F14   &kp F15
    &kp F21   &kp F22   &kp F23   &kp F24   &none           &bt BT_SEL 0   &bt BT_SEL 1   &bt BT_SEL 2   &none     &none
                        &trans    &none     &none          &trans

    >;
};

numpad-L1+L2 {
    bindings = <
    &kp ESC   &kp C_NEXT         &kp C_BRIGHTNESS_INC   &kp C_VOLUME_UP     &kp TAB          &kp KP_DIVIDE     &kp KP_NUMBER_7   &kp KP_NUMBER_8   &kp KP_NUMBER_9   &kp KP_PLUS
    &tog 6    &kp C_PREVIOUS     &kp C_BRIGHTNESS_DEC   &kp C_VOLUME_DOWN   &kp SPACE        &kp KP_MULTIPLY   &kp KP_NUMBER_4   &kp KP_NUMBER_5   &kp KP_NUMBER_6   &kp KP_ENTER
    &tog 4    &kp C_PLAY_PAUSE   &none                  &kp C_MUTE          &sys_reset       &kp KP_SUBTRACT   &kp KP_NUMBER_1   &kp KP_NUMBER_2   &kp KP_NUMBER_3   &kp KP_DOT
                                                        &trans              &trans           &kp BACKSPACE     &kp KP_N0

    >;
};

mirrorNumpad-R1+R2 {
    bindings = <
    &kp KP_PLUS    &kp KP_N7   &kp KP_N8   &kp KP_N9   &kp KP_DIVIDE          &kp TAB      &kp C_VOLUME_UP     &kp C_BRIGHTNESS_INC   &kp C_NEXT         &kp ESC
    &kp KP_ENTER   &kp KP_N4   &kp KP_N5   &kp KP_N6   &kp KP_MULTIPLY        &kp SPACE    &kp C_VOLUME_DOWN   &kp C_BRIGHTNESS_DEC   &kp C_PREVIOUS     &tog 6
    &kp KP_DOT     &kp KP_N1   &kp KP_N2   &kp KP_N3   &kp KP_MINUS           &sys_reset   &kp C_MUTE          &none                  &kp C_PLAY_PAUSE   &tog 5
                               &kp KP_N0   &kp BSPC                           &trans       &trans

    >;
};

mouse-L2+R1 {
    bindings = <
    &msc SCRL_LEFT   &msc SCRL_DOWN   &mmv MOVE_UP     &msc SCRL_UP      &msc SCRL_RIGHT        &msc SCRL_LEFT   &msc SCRL_DOWN   &mmv MOVE_UP     &msc SCRL_UP      &msc SCRL_RIGHT
    &kp ESC          &mmv MOVE_LEFT   &mmv MOVE_DOWN   &mmv MOVE_RIGHT   &kp LEFT_SHIFT         &kp LEFT_SHIFT   &mmv MOVE_LEFT   &mmv MOVE_DOWN   &mmv MOVE_RIGHT   &kp ESC
    &tog 6           &kp LEFT_ALT     &mkp MB5          &mkp MB4          &mkp MCLK              &mkp MCLK        &mkp MB4         &mkp MB5          &kp LEFT_ALT      &tog 6
                                                        &mkp RCLK         &mkp LCLK              &mkp LCLK        &mkp RCLK

    >;
};
};
};
```